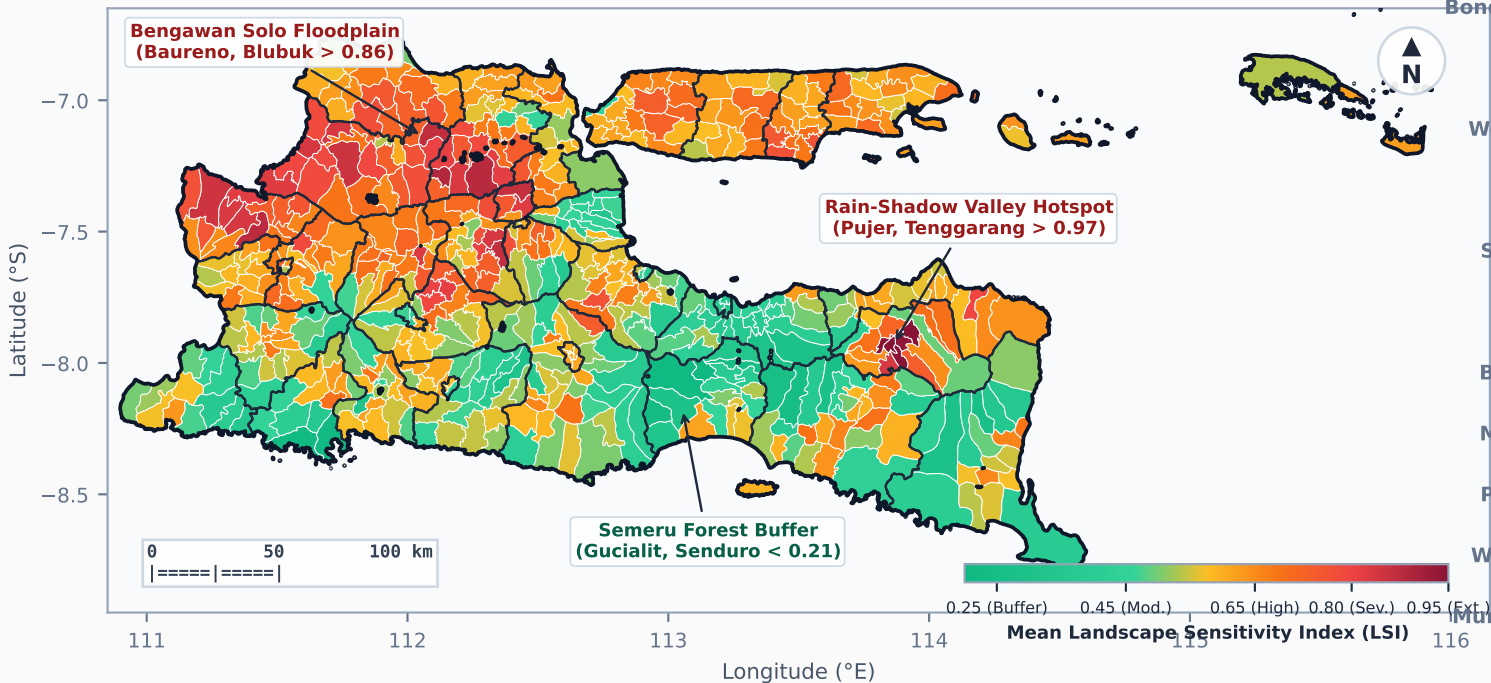


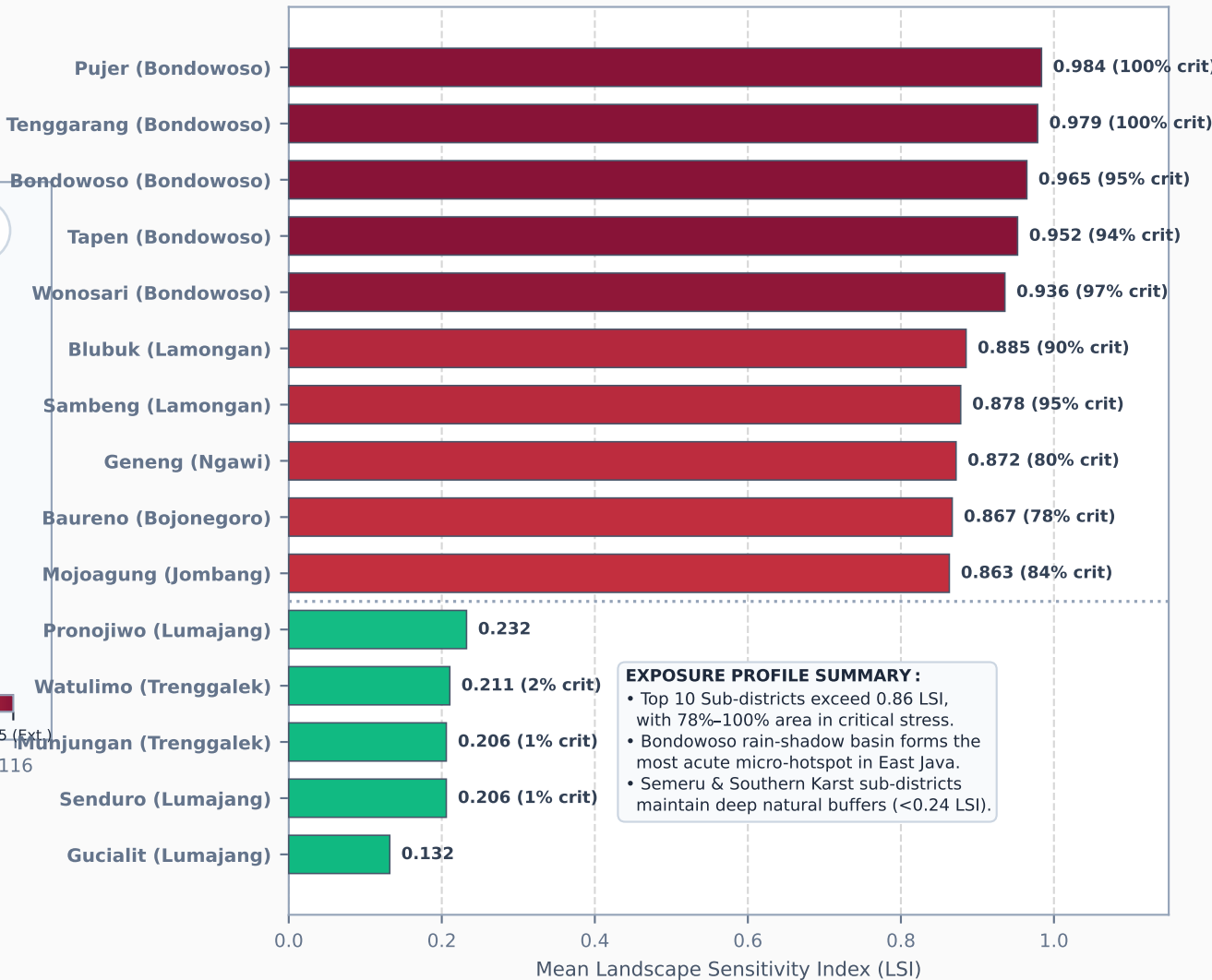
# East Java Hierarchical Landscape Sensitivity Index (H-LSI) by Sub-District

Zonal Statistical Aggregation Across 582 Administrative Kecamatan in 38 Regencies/Cities | 1 km MODIS/CHIRPS/ERA5 Multi-Sensor Evidence | Provincial Mean = 0.585 (Min: 0.132, Max: 0.984)

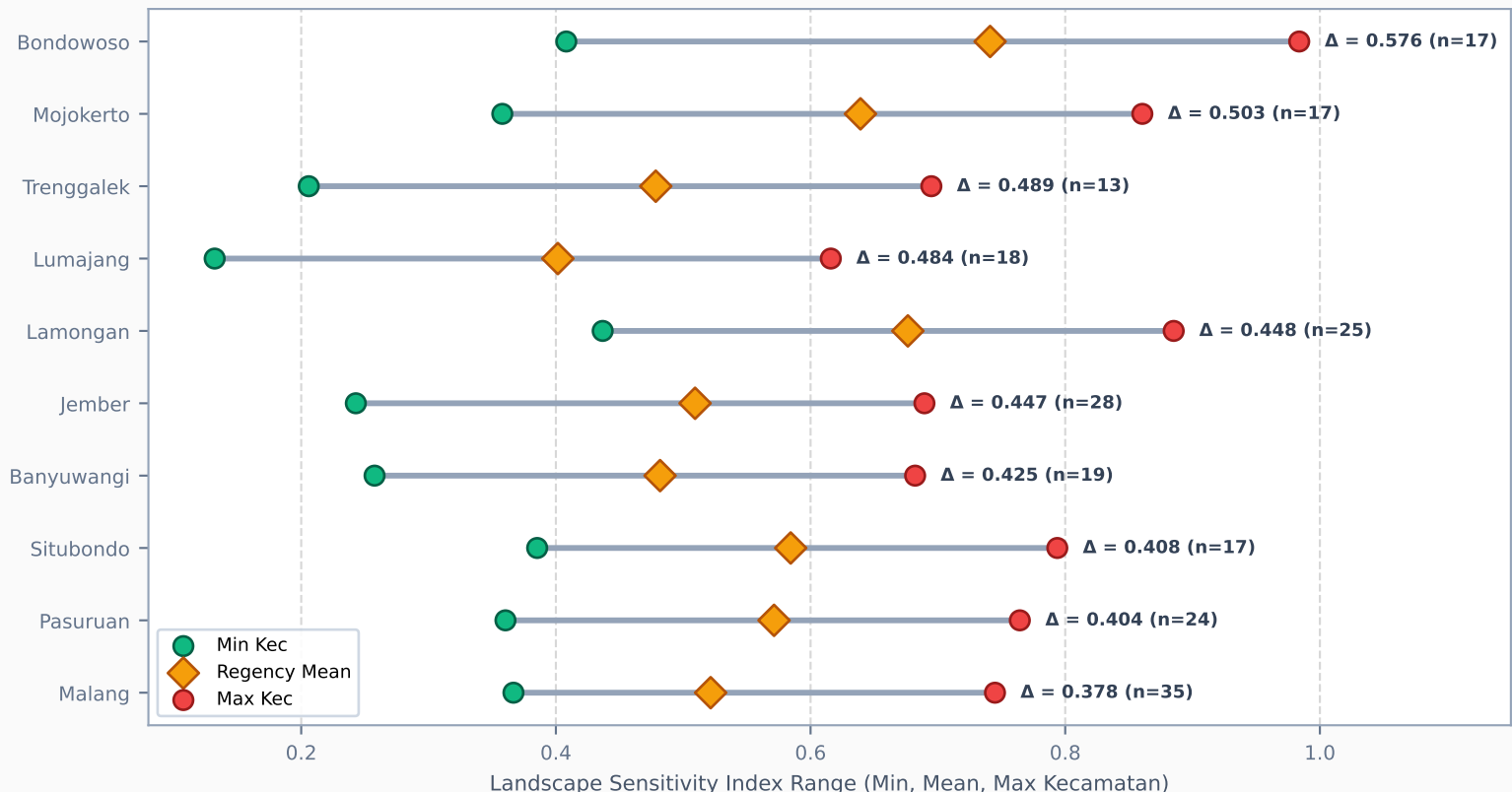
(a) Sub-District Landscape Sensitivity Index (H-LSI, n = 582 Kecamatan)



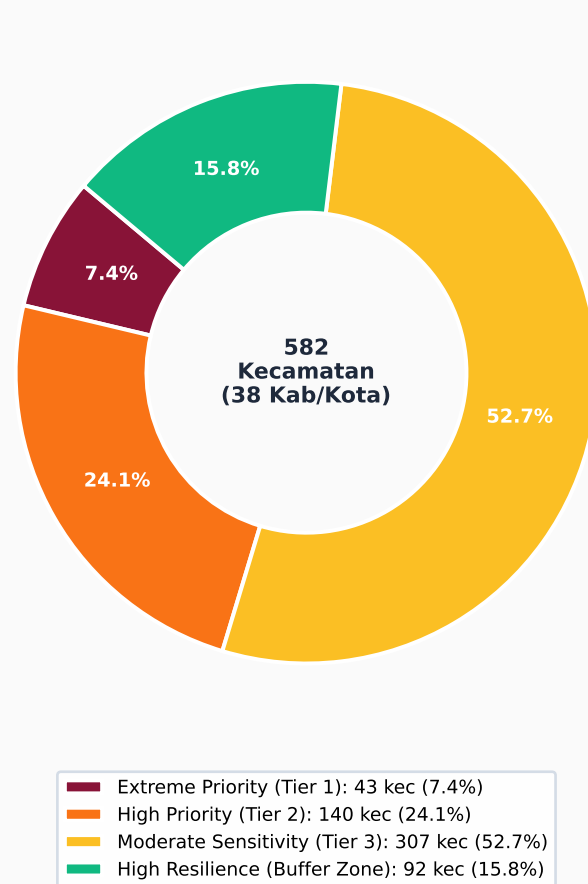
(b) Sub-District Exposure Extremes & Critical Area %



(c) Intra-Regency Climate Risk Disparity (Top 10 High-Variance Kab/Kota)



(d) Administrative Risk Classification Breakdown



**OPERATIONAL POLICY MANDATE (BPBD & BPP LEVEL) :**

- Anti-Ecological Fallacy: Regency-level averages mask massive internal inequalities (e.g. Bondowoso  $\Delta = 0.576$  between Pujer and Ijen).
- Tier 1 Priority (43 Kecamatan): Pre-season budget mobilization for deep boreholes, mobile tankers, and MT-III drought crop shift.
- Natural Buffers (92 Kecamatan): Semeru, Argopuro, and Karst provide hydrological resilience; safeguard watershed forest cover.

**METHODOLOGICAL ARCHITECTURE & FORMULATION :**

- Zonal Spatial Aggregation: Zonal mean, median, IQR, and critical area ( $\% \text{LSI} \geq 0.75$ ) computed per polygon from M5 master LSI.
- Intra-Regency Relative Scoring: Local z-score ( $Z_{ijk} = (\text{LSI}_{i,k} - \mu_k) / \sigma_k$ ) identifies extreme local outlier subdistricts.
- Master Dataset: Full tabular database with provincial & county ranks available in `outputs/m5\_hierarchical\_lsi\_kecamatan.csv`.